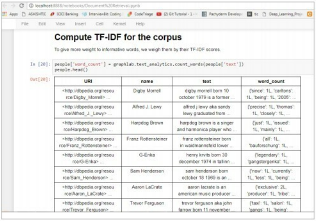

Compute TF-IDF for the corpus

To give more weight to informative words, we weigh them by their TF-IDF scores.

```
In [20]: people['word_count'] = graphlab.text_analytics.count_words(people['text'])
people.head()
```

Out[20]:

URI	name	text	word_count
<http://dbpedia.org/resource/Digby_Morrell> ...	Digby Morrell	digby morrell born 10 october 1979 is a former ...	{'since': 1L, 'carltons': 1L, 'being': 1L, '2006': ...
<http://dbpedia.org/resource/Alfred_J._Lewy> ...	Alfred J. Lewy	alfred j lewy aka sandy lewy graduated from ...	{'precise': 1L, 'thomas': 1L, 'closely': 1L, ...
<http://dbpedia.org/resource/Harpdog_Brown> ...	Harpdog Brown	harpdog brown is a singer and harmonica player who ...	{'just': 1L, 'issued': 1L, 'mainly': 1L, ...
<http://dbpedia.org/resource/Franz_Rottensteiner> ...	Franz Rottensteiner	franz rottensteiner born in waidmannsfeld lower ...	{'all': 1L, 'bauforschung': 1L, ...
<http://dbpedia.org/resource/G-Enka> ...	G-Enka	henry krvits born 30 december 1974 in tallinn ...	{'legendary': 1L, 'gangstergenka': 1L, ...
<http://dbpedia.org/resource/Sam_Henderson> ...	Sam Henderson	sam henderson born october 18 1969 is an ...	{'now': 1L, 'currently': 1L, 'less': 1L, 'being': ...
<http://dbpedia.org/resource/Aaron_LaCrate> ...	Aaron LaCrate	aaron lacrate is an american music producer ...	{'exclusive': 2L, 'producer': 1L, 'tribe': ...
<http://dbpedia.org/resource/Trevor_Ferguson> ...	Trevor Ferguson	trevor ferguson aka john farrow born 11 november ...	{'taxi': 1L, 'salon': 1L, 'gangs': 1L, 'being': ...

Figure 4: Compute TF-IDF for the corpus

4. Next, compute the TF-IDF for the corpus. To give more weight to informative words, we evaluate them based on their TF-IDF scores, as follows:

```
people['word_count'] = graphlab.text_analytics.count_
words(people['text'])
people.head()
```

5. To examine the TF-IDF for the Obama article, give the following commands:

```
obama = people[people['name'] == 'Barack Obama']
obama[['tfidf']].stack('tfidf',new_column_
name=['word','tfidf']).sort('tfidf',ascending=False)
```

Examine the TF-IDF for the Obama article

```
In [23]: obama = people[people['name'] == 'Barack Obama']

In [24]: obama[['tfidf']].stack('tfidf',new_column_name=['word','tfidf']).sort('tfidf',ascendin
```

Out[24]:

word	tfidf
obama	43.2956530721
act	27.678222623
iraq	17.747378588
control	14.8870608452
law	14.7229357618
ordered	14.5333739509
military	13.1159327785
involvement	12.7843852412
response	12.7843852412
democratic	12.4106886973

[273 rows x 2 columns]
Note: Only the head of the SFrame is printed.

Figure 5: TF-IDF for the Obama article

Words with the highest TF-IDF are much more informative. The TF-IDF of the Obama article brings up similar articles that are related to it, like Iraq, Control, etc.

Machine learning is not a new technology. It's been around for years but is gaining popularity only now as many companies have started using it. END

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